

ECON 351 · Midterm Simulation

Practice Test 3

24 Questions | 24 points (1 pt each)

FORMAT (matches your teacher's breakdown):

PART A · Chapter 4 (12 questions)

- 2 Consumer Problems
- 2 Labor-Leisure
- 2 Intertemporal Consumption
- 3 Elasticity / Types of Goods
- 1 Budget Line
- 2 Assumptions / Preferences

PART B · Chapter 9 (12 questions)

- 5 Insurance Problems
- 4 Risk Attitudes
- 2 Certainty Equivalent & Risk Premium
- 1 Diversification

Problems from HW, sample questions, and mock midterm (screenshots).

Use Formula_Sheet.pdf for reference.

PART A

CHAPTER 4

12 questions



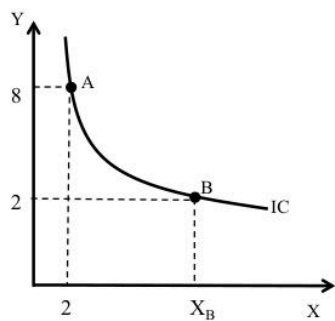
Question 22:

[Show your work]

A consumer has preferences over goods X and Y that can be represented by the following utility function:

$$u(X, Y) = X^6 Y^3.$$

The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

[Q021]

Maria: 16 hrs/day, $w=\$20$, $V=\$80$, $U=c^{0.5} l^{0.5}$. Find I^* , L^* , c^* .
(Course-style)

[Q022]

Same setup, $w=\$30$. Find I^* , L^* , c^* . More or less work?
(Course-style)

Exercise 5 - Intertemporal Consumption

In class, we saw how the baseball player Shohei was offered a peculiar contract. Here, we'll use this example to solve an intertemporal consumption problem.

5.1: Assume the Angels' offer him two contracts: the first pays him 10 million today and 90 million in the future, while the second pays him 42 million today and 54 million in the future. If the interest rate is $r = 0.20$, which contract is cheaper from the Angels' perspective?

5.2: Assume that Shohei, a very patient man, has the following utility function $U(c_1, c_2) = 2c_1^{0.4}c_2^{0.6}$ where c_1 is money for consumption today and c_2 is money for consumption in the future. In plain English, what does it imply that the exponent of c_2 is higher than the exponent of c_1 ? Calculate the ratio of the marginal utilities with MU_{c_1} in the numerator.

5.3: Now let's equalize the MRS to the price ratio, assuming unit prices. That is, $p_1 = p_2 = 1$. Remember to discount p_2 using the interest rates since c_2 is consumption in the future!

5.4: Assume the Angels offer Shohei the first contract. Find his optimum consumption c_1^* and c_2^* under that contract. Once you get them both, plug them into Shohei's utility function to find out how much utility he obtains from that contract.

5.5: Assume instead the Angels offer Shohei the second contract. Find his optimum consumption c_1^* and c_2^* under that contract. Once you get them both, plug them into Shohei's utility function to find out how much utility he obtains from this contract.

5.6 EXTRA: Which contract would Shohei prefer? If, for some reason, the Angels were low on funds today and were also unable to borrow, what type of contract could they offer Shohei such that he accepts it?

Question 24:**[Show your work]**

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 60$ in the first period and $I_2 = 125$ in the second period. The consumer can borrow or save money at the interest rate of 25%, that is, $r=0.25$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

The consumer's inter-temporal budget constraint is

- (A) $q_1 + 0.8q_2 = 160$
- (B) $q_1 + q_2 = 185$
- (C) $q_1 + 1.25q_2 = 216.25$
- (D) $q_1 + 0.75q_2 = 160$
- (E) $q_1 + 0.25q_2 = 200$

Question 25:**[Show your work]**

Consider all the information from Question 24. In the first period, the consumer will optimally purchase

- (A) $q_1 = 0$
- (B) $q_1 = 100$
- (C) $q_1 = 60$
- (D) $q_1 = 80$
- (E) $q_1 = 120$

Exercise 4 - Price Elasticity

Here, we will review the concept of elasticity and go through some examples. Price elasticity of demand is how much quantity demanded changes when the price goes up by 1%. Reminder $E_p = \frac{P}{Q} \frac{\partial Q}{\partial P}$

Take the following demand function $q = 100 - 5p$.

4.1 Calculate the quantity demanded when $p = 5$ and when $p = 10$. What happens to the quantity demanded when the price increases? Given your answer, do you expect the sign of price elasticity to be positive or negative?

4.2 Calculate the price elasticity when $p = 5$. Interpret the elasticity.

4.3 Calculate the quantity demanded when $p = 5.05$.

4.4 Note that $p = 5.05$ corresponds to a 1% price increase with respect to $p = 5$. Compare this % increase with the % change in the quantity demanded. That is, calculate $\Delta Q / Q(p = 5) * 100$. How does this compare to the price elasticity you calculated before?

4.5 Redo points 4.2-4.4 but now calculate the elasticity at $p = 10$ and compare the % change in quantity demanded when $p = 10.1$. How does the % change in quantity compared to the elasticity at $p = 10$?

4.6 EXTRA In the previous questions, you have calculated the elasticity using the formula and also by "brute force". In these cases, the values coincide, but in other cases, there would be a small gap between both methods. Why is that the case, and why are they the same here? (Hard question. Tip: it has to do with the basic concept of a derivative and the curvature of the demand function.)

Question 18:

The price of good A goes up. As a result, the demand for good B shifts to the right. From this we can infer that:

- (A) good A is a normal good.
- (B) good B is a normal good.
- (C) goods A and B are substitutes.
- (D) goods A and B are complements.
- (E) none of the above

Question 19:

If goods X and Y are independent, then their cross-price elasticity is

- (A) zero.
- (B) positive.
- (C) negative.
- (D) $+\infty$.
- (E) $-\infty$.

Question 20:

The cross-price elasticity of demand of goods X and Y is $E_{Q_X, P_Y} = -2$. This means that if the price of good Y increases by 1%, then the quantity demanded for good X will

- (A) increase by 2%
- (B) decreased by 2%
- (C) increase by $\frac{1}{2}\%$
- (D) decreased by $\frac{1}{2}\%$
- (E) not change

Question 21:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are substitutes, and apples are a normal good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

- (A) $Q_A = 100 - 3P_A + 3I - 5P_B$
- (B) $Q_A = 200 - P_A + 2I + 4P_B$
- (C) $Q_A = 110 - 2P_A - 4I + 2P_B$
- (D) $Q_A = 250 - 4P_A - 5I - 3P_B$
- (E) $Q_A = 150 + P_A + I - P_B$

Question 38:

A consulting company researched the market for apples in California, and concluded the following: apples and bananas are complements, and apples are an inferior good.

Define the following:

Q_A : quantity of apples demanded

P_A : price of Apples

P_B : price of bananas

I : income

From the options below, mark the only demand function that can possibly represent the demand for apples in California.

(A) $Q_A = 100 - 3P_A + 3I - 5P_B$

(B) $Q_A = 200 - P_A + 2I + 4P_B$

(C) $Q_A = 110 - 2P_A - 4I + 2P_B$

(D) $Q_A = 250 - 4P_A - 5I - 3P_B$

(E) $Q_A = 150 + P_A + I + P_B$

Question 39:

The demand for apples (Q_A) is given by the function

$$Q_A = 784 - 2P_A - 3I - 4P_O$$

where P_A is the price of apples, I is the income level, and P_O is the price of oranges.

Suppose $P_A = 2$, $I = 100$, and $P_O = 20$.

a) Compute the price elasticity of demand for apples.

b) Compute the income elasticity of demand for apples. Are apples a normal good or inferior good?

c) Compute the cross-price elasticity of demand for apples. Are apples and oranges substitutes or complements?

Question 40:

The income elasticity of demand for good X is -2. Due to a financial crisis, the income of consumers declined by 2%. How did the financial crisis affect the quantity demanded for X?

(A) The quantity demanded increased 2%

(B) The quantity demanded decreased 2%

(C) The quantity demanded increased 4%

(D) The quantity demanded decreased 4%

(E) The quantity demanded did not change

Question 41:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 50%, that is, $r = 0.5$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

What is the optimal choice of q_1 and q_2 ? Is this consumer saving or borrowing? How much?

Question 42:

Consider the following intertemporal consumption problem with one good and two periods. The quantity of the good consumed in period 1 and period 2 are q_1 and q_2 . The price of each unit of the good is \$1 in both periods. The consumer's income is $I_1 = 10$ in the first period and $I_2 = 12$ in the second period. The consumer can borrow or save money at the interest rate of 5%, that is, $r = 0.05$. The consumer's utility function is $u(q_1, q_2) = q_1 q_2$.

Question 11:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples decreases from \$6 to \$3, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

Question 12:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$6, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

Question 13:

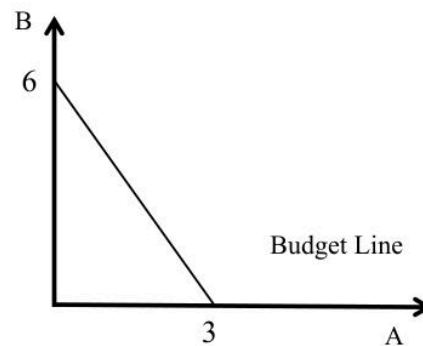
An increase in income, holding prices constant, can be represented as:

- (A) a change in the slope of the budget line.

- (B) a parallel outward shift in the budget line.
- (C) a parallel inward shift in the budget line.
- (D) an outward shift in the budget line with its slope becoming flatter.
- (E) an outward shift in the budget line with its slope becoming steeper.

Question 14:

Novak can consume apples (A) and bananas (B). The graph below represents the consumer's budget line. If we double the price of apples, then the budget line will



- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).
- (E) become steeper or flatter depending on the relationship between prices and income.

[Q003]

Parasite > Indiana Jones > Jurassic Park, but Jurassic Park > Parasite. Violation?

HW1 Ex 2.3

Question 4:

Rafinha has the following utility function for goods X and Y: $u(X, Y) = 6X^{0.5}Y^{0.5}$. When Rafinha consumes $X=9$ and $Y=81$, with X on the horizontal axis, what is the absolute value of the slope of his indifference curve?

- (A) 3.
- (B) 6.
- (C) 9.
- (D) 12.
- (E) 15.

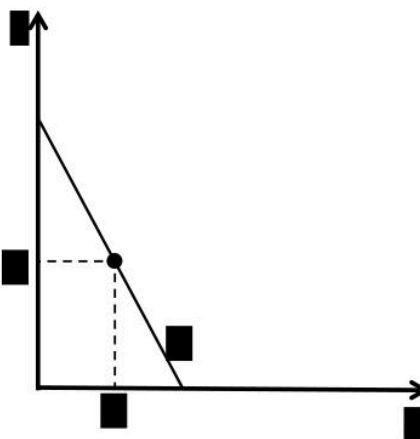
(D) all of the above.

(E) A and C only.

Question 5:

By observing the slope of an indifference curve, we can infer

- (A) that preferences are complete.
- (B) the marginal utility of the good on the vertical axis.
- (C) the marginal rate of substitution of one good for another good.
- (D) the ratio of market prices.
- (E) that preferences are transitive.

**Question 6:**

The magnitude of the slope of an indifference curve is:

- (A) called the marginal rate of substitution.
- (B) equal to the ratio of the total utility of the goods.
- (C) always equal to the ratio of the prices of the goods.

PART B

CHAPTER 9

12 questions

- c) Compute the certainty equivalent of this investment.
- d) Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- (A) $P = 94$
- (B) $P = 130$
- (C) $P = 121$
- (D) $P = 104$
- (E) $P = 100$

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- (A) 23,010
- (B) 45,375
- (C) 74,000
- (D) 32,375
- (E) 15,600

Question 16:

Using the information from the previous question, how much is the fair insurance premium?

- (A) 174,000
- (B) 72,000
- (C) 130,500
- (D) 43,500
- (E) 168,100

Question 19:

Joe's utility from income is given by

$$u(I) = 8\sqrt{I}.$$

He has a house in LA County. With probability $1/4$, the rain will cause a mudslide and damage his house, decreasing its value to \$250,000. With probability $3/4$, no mudslide occurs and the house retains its original \$1,000,000 value. AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Joe is

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

(A) 94

(B) 98

(C) 96

(D) 95

(E) 97

- c) Compute the certainty equivalent of this investment.
- d) Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- (A) $P = 94$
- (B) $P = 130$
- (C) $P = 121$
- (D) $P = 104$
- (E) $P = 100$

Question 15:

Tom has the following utility from income: $u(I) = \sqrt{I}$. He bought a house in Brentwood for \$846,400. With probability 25%, high winds will cause \$174,000 in damages to the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- (A) 23,010
- (B) 45,375
- (C) 74,000
- (D) 32,375
- (E) 15,600

Question 17:

Using the information from Question 15, find Tom's associated risk premium:

- (A) 1,200
- (B) 2,100
- (C) 1,550
- (D) 2,000
- (E) 1,875

maximum insurance premium P that Joe is willing to pay? What is the associated risk premium?

Question 20:

Marcelo has the following utility from income: $u(I) = \sqrt{I}$. He has a \$10,000 car. With probability 25% he will crash his car, and will have to pay \$6,348 to the repair shop to fix it. With probability 75% no crash occurs, and he does not have to repair the car. Progressive Insurance offers to partially insure the car for a price $P=124$. That is,

find Tom's associated risk premium:

- (A) 1,200
- (B) 2,100
- (C) 1,550
- (D) 2,000
- (E) 1,875

Question 18:

Sherman has the following utility from wealth:

$$u(x) = \sqrt{x}.$$

Wealth is the sum of cash and the value of his car. He currently has a Ferrari worth \$230,900 and \$50,000 in cash (risk free). With probability 25%, a paparazzi will cause \$158,400 in damages to the car (in this case, the car value becomes \$72,500). With probability 75% nothing happens, and the car keeps its original value. AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that Sherman is willing to pay? What is the associated risk premium?

Question 20:

Marcelo has the following utility from income: $u(I) = \sqrt{I}$. He has a \$10,000 car. With probability 25% he will crash his car, and will have to pay \$6,348 to the repair shop to fix it. With probability 75% no crash occurs, and he does not have to repair the car. Progressive Insurance offers to partially insure the car for a price $P=124$. That is, Marcelo pays insurance premium $P=124$, and in case of a crash, Progressive will only pay for half of the repair cost (in case of a crash, Progressive will only pay \$3,174 to Marcelo). If Marcelo buys this partial insurance plan, his expected utility will be:

- (A) 94
- (B) 98
- (C) 96
- (D) 95
- (E) 97

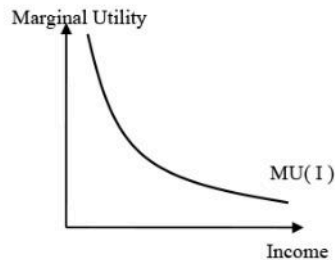
Question 25:

[Show your work] A consumer has the following utility from income: $u(I) = \sqrt{I}$. He bought a car for \$10,000. With probability 75%, a crash will cause \$3,600 in damages to the car (in this case, the car's value becomes \$6,400). AAA offers to fully insure the car for a price P . What is the maximum insurance premium P that the consumer is willing to pay?

- (A) 2,725
- (B) 2,850
- (C) 2,750
- (D) 2,825
- (E) 2,775

Exercise 2 - Attitudes towards risk and fair bets

The following graph represents Mike's **marginal utility** of income (not the utility but the marginal utility):



We know that Mike's income I is greater than \$20, but we do not know his exact income. Alejandro offers Mike the following bet: Alejandro will roll a 6 sided dice (all numbers equally likely). If it is 1 or 2, Mike pays Alejandro \$20. Otherwise, if it is 3, 4, 5 or 6, Alejandro pays Mike \$10.

a) Is this a fair bet?

b) Will Mike accept or reject the bet? Does your answer depend on Mike's initial wealth? (assuming he can always pay the \$20 if he loses)

- a) Compute the expected value of each investment option.
- b) Compute the expected utility of each investment option. Find the best (and worse) investment option for Jim.

Question 3:

Pedro Carroll has the following utility from income: $u(I) = \sqrt{I}$. Currently, he has an income $I = \$169$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Seahawks to win the championship game. If he bets all his money and the Seahawks win, then he ends up with an income $I = \$400$. If he bets all his money and the Seahawks lose, then he ends up with nothing, $I = \$0$. If he does not bet, he keeps his \$169 for sure.

The Los Angeles Seahawks will win the championship game with probability P , and will lose the game with probability $1-P$.

Compute the minimum winning probability P such that Pedro would be willing to bet all his money on the Seahawks.

- (A) $P = 50\%$
- (B) $P = 65\%$
- (C) $P = 70\%$
- (D) $P = 55\%$
- (E) $P = 60\%$

Question 4:

Cam has the following utility from income: $u(I) = \sqrt{I}$. Currently he has an income $I = \$36$ in cash (no risk). He is deciding whether or not to bet all his money on the Los Angeles Panthers to win the championship game. If he bets all his money and

the Panthers win, then he ends up with an income I_{Win} . If he bets all his money and the Panthers lose, then he ends up with nothing, $I_{Lose} = \$0$. If he does not bet, he keeps his \$36 for sure.

The Los Angeles Panthers will win the championship game with probability 75%, and will lose the game with probability 25%.

Compute the minimum winning prize I_{Win} such that Cam would be willing to bet all his money on the Panthers.

- (A) $I_{Win} = \$36$
- (B) $I_{Win} = \$48$
- (C) $I_{Win} = \$56$
- (D) $I_{Win} = \$64$
- (E) $I_{Win} = \$72$

**Question 6:**

Anna's utility of income function is $U(I) = I$, while Elsa's utility of income function is $U(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

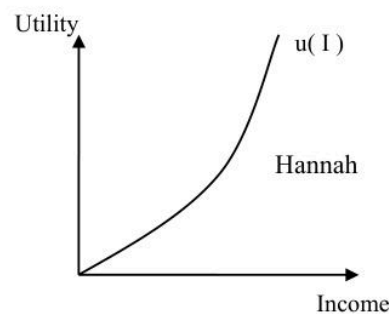
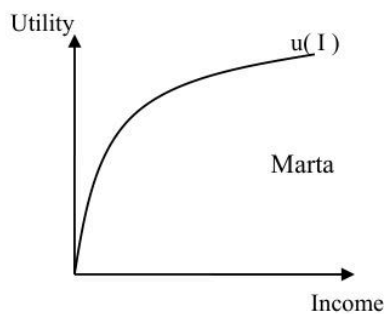
Question 7:

Anna's marginal utility of income function is $MU(I) = I$, while Elsa's marginal utility of income function is $MU(I) = I^{1.5}$. Therefore, Anna is _____ and Elsa is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 8:

The following graphs represent the utility function of consumers Marta and Hannah:

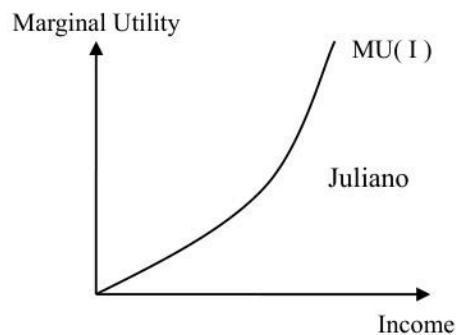
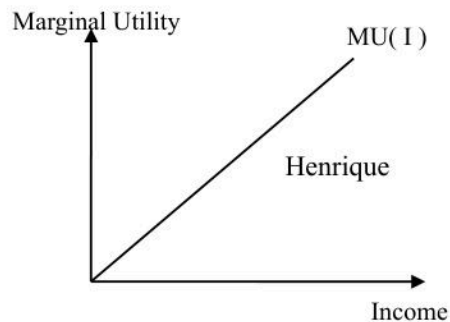


Using the information from the graphs,

- (A) Both Marta and Hannah are risk loving
- (B) Both Marta and Hannah are risk averse
- (C) Marta is risk averse; Hannah is risk loving
- (D) Marta is risk loving; Hannah is risk averse
- (E) Not enough information from the graphs

Question 9:

The following graphs represent the marginal utility of income functions of consumers Henrique and Juliano.



Using the information from the graphs, Henrique is _____ and Juliano is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk loving, risk loving.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Question 10:

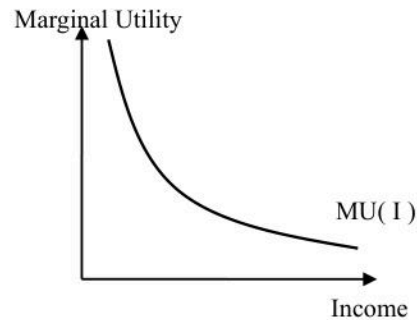
Mike has a constant marginal utility of income. We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough

- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 11:

The following graph represents Mike's marginal utility of income:



We know that Mike's income I is greater than \$10, but we do not know his exact income. Adam offers Mike the following bet: Adam will throw a fair coin (that is, heads and tails have equal probability). If it is heads, Mike pays Adam \$10. If it is tails, Adam pays Mike \$10. Mike will

- (A) reject the bet only if I is high enough
- (B) reject the bet, independently of I
- (C) accept the bet only if I is high enough
- (D) accept the bet, independently of I
- (E) always be indifferent between accepting and rejecting the bet

Question 28:

Firmino is a big fan of a soccer team called The Reds. Everyone knows that there is a 75% chance that the The Reds will win the next tournament, and a 25% chance that they will lose. Firmino offers the following bet to each one of his friends: if The Reds win, then the friend must pay \$1 to Firmino. If The Reds lose, then Firmino will pay \$3 to the friend.

The friends Mohamed and Sadio (who are very rich) are each considering whether to accept the bet. Mohamed's utility of income function is

$$U(I) = I^{0.7}$$

while Sadio's utility of income function is

$$U(I) = I^{2.3}$$

Therefore,

- (A) Both Mohamed and Sadio will reject the bet
- (B) Both Mohamed and Sadio will accept the bet
- (C) Mohamed will accept and Sadio will reject the bet
- (D) Mohamed will reject and Sadio will accept the bet
- (E) Both Mohamed and Sadio are indifferent between accepting and rejecting the bet

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negatively correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Question 30:

Mirabel's marginal utility of income function is $MU(I) = 9$, while Bruno's marginal utility of income function is $MU(I) = \frac{3}{I}$. Therefore, Mirabel is _____ and Bruno is _____.

- (A) risk neutral, risk loving.
- (B) risk averse, risk averse.
- (C) risk neutral, risk averse.
- (D) risk averse, risk loving.
- (E) risk loving, risk averse.

Exercise 1 - Risk Aversion and Certainty Equivalent

Jackie owns shares of a company. With probability 80%, the value of the investment will go up to $x = \$200$, and with probability 20% the value of the investment will go down to $x = \$50$. Jackie's utility from income is given by the function

$$u(x) = \sqrt{x}$$

- a) Compute the expected value of the investment and Jackie's expected utility from holding this investment.
- b) Compute the certainty equivalent of this investment. In other words, the certain amount that makes Jackie indifferent between holding the investment or selling it for that certain amount (the certain amount provides the same expected utility as the investment).
- c) Compute the risk premium associated with this investment.

Question 13:

Cuban is an investor who used all his income to buy a startup company called Shark Aquarium. With probability 50% the value of the company will increase to \$144. With probability 50% the value of the company will go down to \$64. Cuban's utility from income is

$$u(I) = \sqrt{I}.$$

He has no other source of income. Compute the minimum price P such that Cuban would be willing to sell all his shares of Shark Aquarium.

- (A) $P = 94$
- (B) $P = 130$
- (C) $P = 121$
- (D) $P = 104$
- (E) $P = 100$

the house (in this case, the house value becomes \$672,400). AAA offers to fully insure the house for a price P . What is the maximum insurance premium P that Tom is willing to pay?

- (A) 23,010
- (B) 45,375
- (C) 74,000
- (D) 32,375
- (E) 15,600

Question 12:

A consumer owns shares of a company. With probability 80% the value of the investment will go up to $x = \$144$, and with probability 20% the value of the investment will go down to $x = \$81$. The consumer's utility from income is given by the function

$$u(x) = \sqrt{x}.$$

- a) Compute the consumer's expected utility from holding this investment.
- b) Compute the expected value of this investment.

Question 14:

Using the information from the previous question, compute the associated risk premium.

- (A) 3
- (B) 1
- (C) 7
- (D) 10
- (E) 4

Question 21:

Arthur has the following utility from income:

$$u(I) = \sqrt{I}.$$

He has a collection of rare comic books that is worth \$10,000. With probability 50% a silverfish invasion will destroy all his books, bringing the value of his collection to zero. With probability 50% nothing occurs, and his collection keeps its full value. A company called Insect Invasion Insurance (III) offers to partially insure the collection for a price $P = \$1,900$. That is, Arthur first pays the insurance premium $P = \$1,900$; and then, in case of a silverfish invasion, III will send to Arthur a \$2,044 check. If Arthur buys this partial insurance plan, then his expected utility will be _____. Therefore, Arthur _____.

[Be careful with your computation. Keep as many decimal places as possible along all the steps of the problem. Once you are done with all computations, round your final answer to the nearest integer. It should be clear which one.]

- (A) 53 ; should buy the insurance
- (B) 51 ; should buy the insurance
- (C) 51 ; should not buy the insurance
- (D) 53 ; should not buy the insurance
- (E) None of the above, because Arthur is indifferent between buying the insurance or not.

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that it

will rain a lot, in which case the value of her investment in Company A will go up to \$100 while her investment in Company B will go up to \$90. There is a 50% chance of a drought, in which case the value of her investment in Company A will go down to \$80, while her investment in Company B will go down to \$70.

Nan has her wealth invested in Company C and in Company D. Company C is an airline company, while Company D is an oil company. Their value can go up or down depending on the international price of oil. There is a 50% chance that the price of oil will go up, in which case her investment in Company D will go up to \$100 while her investment in Company C will go down to \$70. There is a 50% chance that the price of oil will go down, in which case the value of her investment in Company D will go down to \$80, while her investment in Company C will go up to \$90.

- a) Are the values of Company A and Company B positively or negatively correlated?
- b) Are the values of Company C and Company D positively or negatively correlated?
- c) Who has the riskier investment portfolio, Feng or Nan?

Question 23:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$120	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

Question 24:

Mark wants to buy a new car to become a rideshare driver. He can buy a regular car or an electric car. His profit depends on the type of car he purchases and on whether the future price of gas will be high or low. The following table summarizes his profit for each type of car and for the different gas prices.

	Low Prices	High Prices
Regular Car	\$110	\$80
Electric Car	\$102	\$90

There is a 50/50 chance that the gas price will be low or high. Mark wants to maximize his expected profit.

- a) Should Mark purchase a Regular Car or an Electric Car?

b) _____
